

CB = 50 marks

1. a) $f_s/N \rightarrow$ integers $\rightarrow$ No T possible (2)
- b) First window then pad zero (2)
- c) $\tilde{x}(n) = x(n)_N, -M \leq n \leq N-1 \rightarrow \tilde{x}(n-k) = x(n-k)_N, -M \leq n-k \leq N-1$
 $y(n) = \tilde{x}(n) * h(n) = x(n) \odot h(n)$ (2)
- d) No difference, both repeat after 18ms $\rightarrow$ have identical freq. (2)

2. $F_1 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$ (1) $T_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ (1)

Rest zeros only diagonals

$F_2 = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 \end{bmatrix}$ (1) $T_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ (1)

Rest zeros

$F_3 = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 \end{bmatrix}$ (1)

$DH = F_1 T_1^* F_2 T_2^* F_3$ (1)

Butterfly! Same as DIF with twiddle factors conjugated (3)

Calculates $N \cdot \text{IDFT}\{x[n]\}$ (1)

3. $z_1 = \text{conjugate of } z_0$
 $z_2 = 0.8839 + j0.8839 = 1.25 e^{j\pi/4}$ (3)
 $z_3 = \downarrow \text{conj of } z_2$

4. $K_2 = 0.25, K_1 = -0.8 \times \alpha_1 = 3, \alpha_2 = 5, \alpha_3 = 4.25 \rightarrow$ Lattice ladder structure (2)

5. $\Omega_p = \frac{2}{T} \times 0.3279$
 $\Omega_s = \frac{2}{T} \times 0.4327$ (2) Prewarping

freq. $0, 0.2\pi, 0.26\pi, \pi$ (1) ; $N_c = 4$ (2)
 $N_B = 7$ (2)

Better design Chebyshev (1)

6. $y[n] = x[n], n = 3K$ with steps (4)

7. $|A_e^2(j\omega)| \rightarrow$ Type I or II (2) $|A_e^3(j\omega)| \rightarrow$ Type I (with resonance) (2)

• DFT = $\begin{bmatrix} 6 \\ 8+4j \\ -2 \\ 8-4j \end{bmatrix}$ Mod-4 red $\begin{bmatrix} 5 \\ 0 \\ -3 \\ 4 \end{bmatrix}$

$8 = ?^0$ $6 + (-2) = 4$ (4)

• Circ. conv. (any method) $\begin{bmatrix} -4 \\ 20 \\ 0 \\ -19 \end{bmatrix}$; $(?^1) = -3 = 3$ Abs. value

(4) $(?^3) = \frac{20}{2} = 10$

$(?^2) = \frac{10}{2} = 5$

Q1: $?^0 = 8$	Q4: $?^2 = 5$	Q7: $?^4 = 4$
Q2: $?^3 = 10$	Q5: $?^0 = 8$	Marking Scheme: 8
Q3: $?^1 = 3$	Q6: $?^4 = 4$	Total = 50

OB 30 Marks

1. (a) $y_1(n) = x(n) \cdot w_2(n)$ $\rightarrow 0.42 - 0.5 \cos\left(\frac{2\pi n}{N}\right) + 0.08 \cos\left(\frac{4\pi n}{N}\right)$

(4) Same as Blackmann with $(N-1)$ replaced by N

(b) DFT first $\rightarrow$ computing $\downarrow$, saving window coeff (no need)

(c) Hamming, Hanning (2)

2. (a) $[1, h, 1] \rightarrow$ DTFT $\rightarrow$ Simplify $\Rightarrow$ use $H(wn) = 0$

$h_1 = 1.6617$ (6)

(b) DC Gain = 3.6617 (2)

(c) Linear ph $\rightarrow$ as coeff are symmetrical (2)

3. (a) Use: DFT in samples of DTFT $\hat{X}(k) = \begin{cases} 2X(k/2); & k \rightarrow \text{even} \\ 0; & k \text{ odd.} \end{cases}$

Show steps (6)

(b) we want $X(k)G(k) = 2X(k)H(2k)$

$\therefore g(n) = 2[h(n) + h(n+N)]$ (3) $\rightarrow$ System A: Time axis $\times$ multiplies by 2

System B $y[k] = w[k/2]$ even $k \rightarrow$ we need

$y[n] = \begin{cases} w[n], & 0 \leq n \leq N-1 \\ w[n-N], & N \leq n \leq 2N-1 \end{cases}$

System B generates $2N$ -pt seq. by repeating $w[n]$ (3)